

Novelty-Aware Agentic Retrieval: Comparing Research Contributions Through Structured Multi-Step Reasoning

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Abstract

Scientific literature search is an information retrieval (IR) task in which ranked lists are insufficient: a researcher entering a new area needs to know not only *which* papers are relevant, but how they relate: where they overlap, how they differ, and what problem–method combinations are absent. Standard retrieval-augmented generation (RAG) summarizes documents independently, discarding exactly this comparative signal. We present the *Novelty-Aware Research Agent*, a prototype agentic retrieval system that layers structured multi-step reasoning on a RAG pipeline through six typed-contract components: query analysis, a ReAct-style retrieval loop, relevance ranking, schema-guided contribution extraction, a three-pass comparison agent, and answer generation. Beyond returning relevant papers, it produces structured comparison artifacts: per-paper contribution records, paper-level overlaps, and a problem $\times$ method gap matrix. On a 100-paper corpus, our central result is that the system supports five structured comparison capabilities that a standard RAG baseline supports none of, while remaining query-sensitive. Across the three main queries, no paper appears in all three top-5 sets and the mean pairwise Jaccard similarity is 0.12; in an extended seven-query evaluation, the same pattern holds across ten total queries, with mean pairwise Jaccard similarity of 0.115 and 18 of 29 distinct retrieved papers appearing in only one query. Under author-assigned graded relevance, the ranker attains mean Precision@5 of 1.000 and nDCG@5 of 0.752 on the three main queries, ahead of BM25, dense, and hybrid retrieval; over all ten queries, Precision@5 remains high but non-saturated at 0.980, with nDCG@5 of 0.739. Schema compliance is 86.7% on the main queries and 84.0% over the ten-query set, and a validation of 20 sampled empty gap-matrix cells yields gap precision of 0.600. We discuss the latency–structure trade-off inherent to agentic retrieval and identify corpus scale, author-assigned relevance labels, and limited independent evaluation as the primary limitations of the prototype.

CCS Concepts

• **Information systems** $\rightarrow$ **Information retrieval**; *Retrieval models and ranking*; • **Computing methodologies** $\rightarrow$ *Artificial intelligence*.

Keywords

agentic information retrieval, retrieval-augmented generation, ReAct, constrained decoding, scientific literature search, contribution comparison, corpus-level gap analysis, LLM agents

1 Introduction

The integration of large language model (LLM) agents into information retrieval has reshaped how users find and consume information. Agentic retrieval systems, those combining tool use, memory, reasoning, and planning, can decompose queries, retrieve evidence, and synthesize responses in ways that go beyond returning a ranked list of documents. Yet for one common and important IR task, scientific literature search, even agentic systems tend to fall back on a familiar pattern: retrieve relevant documents, then summarize each one independently.

This pattern is inadequate for the underlying user need. A researcher entering a new area does not primarily want a list of papers or a set of disconnected summaries. They want to understand the *structure* of a body of work: which papers address the same problem, which propose genuinely different methods, and crucially, what problem–method combinations have *not* yet been explored. These are comparative and corpus-level questions. A retrieval system that summarizes documents independently discards exactly the signal needed to answer them. Concretely, the payoff of answering them is that a researcher can see, in one structured view, which retrieved papers converge and which problem–method combinations remain open, rather than reconstructing that map by reading and cross-referencing a ranked list by hand.

We frame this as a novelty-aware retrieval problem and present the *Novelty-Aware Research Agent*, a prototype agentic retrieval system that layers structured comparison reasoning on top of a retrieval-augmented generation (RAG) pipeline. Rather than asking “what does this paper say?” the system asks “what does this retrieved set collectively cover, where does it converge, and what is absent?”

The system retrieves papers from a domain corpus, ranks them by query relevance, extracts a structured contribution record for each, and then runs a dedicated comparison stage that identifies overlaps, differentiating aspects, and a problem $\times$ method gap matrix. The final output is not a single prose summary but a set of typed, auditable artifacts that a user (or a downstream system) can inspect and act on.

Contributions.

- (1) A six-component agentic retrieval pipeline with typed inter-component contracts, combining a ReAct-style retrieval loop, relevance ranking, schema-guided extraction, and a multi-pass comparison stage.
- (2) A three-pass Comparison Agent that operates on structured contribution records rather than raw retrieved text, producing paper-level overlaps, per-paper differentiation, and a deterministic problem $\times$ method gap matrix.

- (3) An evaluation on a 100-paper agentic AI and retrieval corpus, including query-sensitivity analysis, graded-relevance retrieval metrics (Precision@5, nDCG@5, Recall@5, MRR), schema-compliance failure analysis, a ReAct sparse-query stress test, deterministic gap-matrix construction, a baseline comparison, BM25/dense/hybrid retriever comparison, component ablation, deterministic gap validation, gap error analysis, a qualitative case study, an indicative single-rater external usefulness assessment collected on an earlier system version, and an extended seven-query evaluation.
- (4) A working open-source prototype implementation with a web interface that streams the six-stage retrieval process in real time.¹

We position this as a prototype system paper. The contribution is the retrieval architecture and the comparison artifacts it produces, not a large-scale benchmark; we are explicit throughout about the scale and evaluation limitations of the current corpus.

2 Related Work

Agentic IR and RAG. LLM agents extend retrieval beyond single-shot ranking through query decomposition, iterative search, and tool use. Multi-agent frameworks such as AutoGen [4] and MetaGPT [5] coordinate agents for complex tasks, AgentBench [8] evaluates agent behavior across environments, and a recent survey [12] reviews agent architectures. These establish the agentic paradigm but do not target the IR task of comparing retrieved documents at the level of their contributions. Our system builds on retrieval-augmented generation (RAG) [1], including its adaptive and self-critical variants Self-RAG [10] and corrective retrieval [11], using FAISS exact nearest-neighbor search as a substrate. It departs from typical RAG output: where standard RAG concatenates passages into one response, our system retains each retrieved paper as a distinct unit, extracts a structured record, and reasons across records.

Reformulation, structured extraction, and comparison. Effective retrieval often depends on reformulating the query. ReAct [2] interleaves reasoning and action in a Thought $\rightarrow$ Action $\rightarrow$ Observation loop; we apply this at the retrieval stage, placing reformulation under explicit agent control rather than a single fixed embedding. JSON schema-guided decoding [3] constrains generation to structurally valid output; we use it to extract a fixed four-field contribution record per paper, so downstream comparison operates over uniform inputs. Chain-of-thought prompting [13] and deployment platforms such as OpenAgents [14] are adjacent reasoning and agent-deployment work covered by our corpus. We are not aware of prior agentic retrieval systems that combine structured contribution extraction, cross-paper comparison, and deterministic problem-method gap-matrix construction in a single literature-comparison pipeline. Table 1 positions our system relative to representative prior work across five capability dimensions.

Automated literature review and gap analysis. A parallel line of work automates systematic literature reviews (SLRs) with LLM agents. Sami et al. [21] propose a multi-agent system that automates the full SLR workflow, including agents that retrieve, filter, and summarize papers and surface trends and gaps, and Moses

Algorithm 1 Novelty-Aware Agentic Retrieval

Require: query q , corpus C , min-papers k , top- n N

Ensure: structured comparison report R

```

1:  $Q \leftarrow \text{QUERYANALYZER}(q)$  ▷  $T=0.3$ 
2:  $iter \leftarrow 0$ 
3: while  $iter < 3$  do
4:    $P \leftarrow \text{FAISS.SEARCH}(Q.\text{queries}[iter], C)$ 
5:   if  $|\text{UNIQUE}(P)| \geq k$  then
6:     break ▷ ReAct STOP
7:   end if
8:    $q' \leftarrow \text{LLM.REASON}(q, P)$  ▷ ReAct REFINE
9:    $Q.\text{queries}.\text{APPEND}(q'); iter++$ 
10: end while
11:  $P_n \leftarrow \text{RANKER}(P, q, N)$  ▷  $T=0.1$ 
12:  $E \leftarrow \text{EXTRACTOR}(P_n)$  ▷ schema;  $T=0.1$ 
13:  $O \leftarrow \text{OVERLAPPASS}(E)$  ▷  $T=0.2$ 
14:  $D \leftarrow \text{DIFFERENTIATIONPASS}(E)$  ▷  $T=0.2$ 
15:  $G \leftarrow \text{GAPMATRIX}(E)$  ▷ deterministic
16:  $R \leftarrow \text{ANSWERGENERATOR}(q, E, O, D, G)$  ▷  $T=0.7$ 
17: return  $R$ 
```

et al. [22] introduce a gap-aware agentic workflow that combines structured synthesis, knowledge-graph modeling, and perspective-guided questioning to identify gaps in coverage, reasoning, or evidence through graph traversal and contrastive retrieval. Our system shares the multi-agent and gap-surfacing motivation but differs in target and mechanism: rather than generating a written review narrative, it treats the retrieved set as structured data and constructs a deterministic problem $\times$ method gap matrix over typed contribution records, so that the gap decision is a reproducible structural computation rather than generated text.

3 System Architecture

The system comprises six components arranged in a sequential agentic retrieval pipeline with defined input/output contracts. Figure 1 shows the data flow and Algorithm 1 formalizes the procedure. Lines 3–10 implement the ReAct retrieval loop; lines 13–15 implement the three comparison passes.

3.1 Query Analyzer

The entry point decomposes the user query into focused retrieval intents and 2–4 reformulated queries, operating at temperature $T=0.3$ under JSON schema constraints. Reformulation is included to support sparse or underspecified retrieval settings, with a modest measurable benefit on the present 100-paper corpus (Section 5).

3.2 Retriever (ReAct Loop)

The retriever performs FAISS IndexFlatL2 search over sentence-transformer embeddings with ReAct-style refinement. If fewer than three unique papers meet an $L2 < 400$ distance threshold, the agent emits a Thought explaining the shortfall and an Action query reformulation, retrying up to three iterations. In the main broad-query experiments retrieval succeeded in one iteration on every run. We therefore additionally conduct a sparse-query stress test

¹Code: <https://github.com/52147/Novelty-Aware-Research-Agent> Live demo: <https://debrahl1-novelty-aware-research-agent.hf.space>

Table 1: Positioning of the proposed system relative to representative prior work.

System / Work	RAG	Agent Loop	Structured Extraction	Cross-Paper Comparison	Problem-Method Gap Matrix
RAG [1]	Yes	No	No	No	No
ReAct [2]	Tool-dep.	Yes	No	No	No
Self-RAG [10]	Yes	Yes	No	No	No
AutoGen [4]	No	Yes	No	No	No
MetaGPT [5]	No	Yes	No	No	No
AgentBench [8]	No	No	No	No	No
SLR multi-agent systems [21]	Yes	Yes	Partial	Yes	No
Gap-aware workflows [22]	Yes	Yes	Yes	Yes	Partial
Novelty-Aware Research Agent (ours)	Yes	Yes	Yes	Yes	Yes

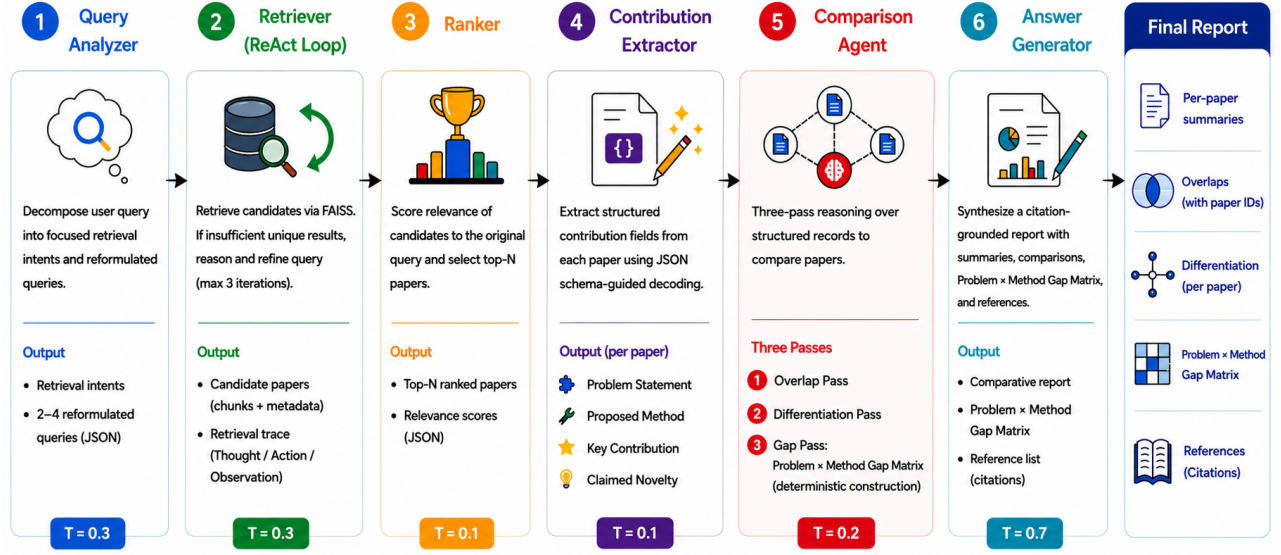


Figure 1: End-to-end agentic retrieval pipeline. A user query is decomposed and reformulated, candidate papers are retrieved and ranked, structured contribution records are extracted under a schema, three comparison passes produce overlaps / differentiation / a gap matrix, and a final citation-grounded report is generated.

(Section 5.7), in which the loop activates on three of five strict-threshold queries and recovers sufficient coverage in one of the three triggered cases.

3.3 Ranker

The ranker scores each retrieved candidate for query relevance at $T=0.1$ and returns the top- N . Low temperature yields consistent, near-deterministic scoring. As Section 5 shows, the ranker selects substantially different paper sets across queries, evidence that scoring is query-sensitive rather than returning a fixed corpus ordering.

3.4 Contribution Extractor

For each top-ranked paper, schema-guided decoding enforces a four-field record: *Problem Statement*, *Proposed Method*, *Key Contribution*, and *Claimed Novelty*, at $T=0.1$. We define schema compliance as the fraction of papers for which all four fields are non-empty and non-placeholder. Structural validity is guaranteed by decoding; field

completeness depends on the input paper, which is why compliance can fall below 100% for atypically structured papers.

3.5 Comparison Agent

The comparison agent is the core contribution and operates on structured records rather than raw retrieved text, which bounds the reasoning space and makes outputs auditable. It runs three sequential passes:

- **Overlap Pass.** Identifies papers sharing the same problem formulation, dataset, or method family, returning paper identifiers with a shared-element label.
- **Differentiation Pass.** Identifies what each paper does distinctly in method, scope, or claimed contribution.
- **Gap Pass.** Maps retrieved paper identifiers to canonical problem and method labels using a fixed taxonomy; the final problem x method matrix is constructed deterministically, and empty cells are reported as candidate corpus-level gaps within the retrieved paper slice (Section 5.8).

3.6 Answer Generator

The final stage synthesizes a report at $T=0.7$, producing per-paper summaries, a synthesis paragraph, and a citation-grounded reference list. Higher temperature here favors readable natural-language synthesis, in contrast to the precision-oriented extraction and ranking stages.

4 Implementation

4.1 Technology Stack

The system is implemented in Python 3.9 using the OpenAI API [15], with GPT-4o as the primary backbone for the main experiments and GPT-4o-mini and GPT-4.1 used in the cross-model robustness analysis (Section 5.14). The system uses FAISS IndexFlatL2 for exact nearest-neighbor retrieval, Sentence-Transformers (all-MiniLM-L6-v2, 384-dimensional embeddings) for encoding, and Pydantic v2 for typed inter-component contracts. A FastAPI backend streams the six-stage retrieval process to a web interface using server-sent events, allowing users to observe each stage complete in real time.

4.2 Corpus

The evaluation corpus contains 100 papers spanning the agentic AI and retrieval domain: multi-agent frameworks, reasoning techniques, tool-use systems, agent deployment platforms, evaluation and survey work, and a substantial set of retrieval-augmented generation papers. Representative papers include Self-RAG [10], Reflexion [7], CAMEL [6], Generative Agents [9], AgentVerse [16], ReWOO [17], and MetaAgents [18]. Each paper is chunked into abstract, introduction, and conclusion sections, producing 300 FAISS vectors. The corpus defines the system’s entire search space; the system does not retrieve from the open web.

4.3 Temperature Configuration

Temperature is assigned per component following standard practice: low values for precision-critical stages (ranking and extraction at $T=0.1$, comparison at $T=0.2$), moderate values for query analysis and ReAct reasoning ($T=0.3$), deterministic construction for the gap matrix, and a higher value ($T=0.7$) for natural-language synthesis.

5 Experiments and Results

We evaluate using (a) automated retrieval and compliance metrics, (b) graded-relevance retrieval quality (Precision@5, nDCG@5, Recall@5, MRR), (c) schema failure analysis, (d) query-sensitivity analysis, (e) a ReAct sparse-query stress test, (f) deterministic gap-matrix construction, (g) a baseline comparison, (h) BM25/dense/hybrid retriever comparison, (i) component ablation, (j) deterministic gap validation, (k) gap error analysis, (l) a qualitative case study, (m) an indicative single-rater external usefulness assessment collected on an earlier system version, and (n) an extended seven-query evaluation. Three main runs were executed with distinct broad comparison queries on the expanded 100-paper corpus:

- **R1:** “Compare multi-agent LLM frameworks for collaborative reasoning”
- **R2:** “What evaluation methods exist for LLM reasoning agents?”

Table 2: Automated metrics across three runs (100-paper corpus).

Metric	R1	R2	R3	Avg
Corpus size	100	100	100	100
Candidates retrieved	9	10	9	9.3
ReAct iterations	1	1	1	1.0
Schema compliance	80%	80%	100%	86.7%
Overlaps detected	3	3	2	2.7
Differences	5	5	5	5.0
Gaps identified	5	4	5	4.7
Runtime (s)	22.3	23.5	22.9	22.9

- **R3:** “Compare verbal reinforcement and role-playing approaches in LLM agents”

Unless otherwise stated, the main tables report these three primary runs. We additionally report an extended seven-query evaluation (R4–R10) in Section 5.15 to test whether the retrieval behavior remains stable across a wider query set.

5.1 Automated Metrics

Table 2 reports automated metrics. Retrieval returns nine to ten candidates per query in these runs; the ranker reduces them to a top-5 set. Schema compliance averages 86.7%, and the comparison agent produces 2–3 overlaps and 4–5 report-level gaps per run. End-to-end latency averages 23 seconds.

5.2 Query-Sensitive Retrieval

The central retrieval result is that the ranker selects substantially different paper sets across queries (Table 3). No paper appears in all three top-5 result sets. Across the 15 ranked slots the system retrieves 12 distinct papers, 9 of which (75%) are query-exclusive, appearing in only a single run. The mean pairwise Jaccard similarity across the three top-5 sets is 0.12, indicating the retrieved sets are roughly 88% distinct on average; runs R2 and R3 share no papers at all. This indicates the ranking stage responds to query semantics rather than returning a fixed corpus ordering, a necessary property for an agentic retrieval system whose value depends on tailoring the retrieved set to the user’s specific comparative question. Notably, several papers added in the corpus expansion (OpenAgents, ART [19], and RAP [20]) are selected into top-5 sets, confirming that the enlarged corpus actively changes retrieval rather than being ignored. The query-sensitivity statistics indicate that the enlarged corpus does not collapse the system into a fixed retrieval pattern.

5.3 Retrieval Quality

Query-sensitivity shows the retrieved sets *differ* across queries, but not whether they are *good*. To assess ranking quality directly, we labeled the relevance of corpus papers for each of the three main queries on a four-point graded scale (3 = highly relevant, 2 = relevant, 1 = marginal, 0 = not relevant) and computed Precision@5, nDCG@5, Recall@5, and mean reciprocal rank (MRR). Relevance

Table 3: Top-5 papers selected per run on the 100-paper corpus. No paper appears in all three runs.

R1: Multi-agent	R2: Evaluation	R3: Verbal RL
AgentVerse	AgentBench	Reflexion
AutoGen	AgentSurvey	CAMEL
AgentSurvey	ART	AgentVerse
AgentBench	Chain-of-Thought	Gen. Agents
OpenAgents	RAP	Inner Monologue

Table 4: Retrieval quality on the three main queries (author-assigned graded relevance). Precision@5 and MRR are saturated; nDCG@5 and Recall@5 are the more discriminative measures.

Query	P@5	nDCG@5	Recall@5	MRR
R1 (multi-agent)	1.000	0.719	0.455	1.000
R2 (evaluation)	1.000	0.588	0.500	1.000
R3 (verbal RL)	1.000	0.950	0.625	1.000
Mean	1.000	0.752	0.527	1.000

labels are author-assigned; we report this as a limitation in Section 7, and treat the labels as a small-scale, single-annotator gold standard rather than a benchmark-grade resource.

Table 4 reports the results. Precision@5 is 1.0 on all three queries: every paper the ranker placed in the top-5 was judged at least marginally relevant. We read this cautiously: perfect precision on author-assigned labels reflects that the ranker avoids clearly off-topic papers rather than that retrieval is solved. The more informative signals are nDCG@5 and Recall@5. Mean nDCG@5 is 0.752, with the evaluation query (R2) lowest at 0.588: although all five retrieved papers were relevant, their *ordering* did not match the ideal graded ranking, indicating the ranker captures relevance better than fine-grained priority. Mean Recall@5 is 0.527, meaning a top-5 cutoff recovers roughly half of the papers labeled relevant in the corpus, expected given that several queries have more than five relevant papers, and a useful characterization of the precision-recall trade-off at this cutoff. MRR is 1.0 throughout: the top-ranked paper was always relevant.

5.4 Retriever Comparison

To compare the ranker against simpler retrieval alternatives, we evaluated BM25, dense retrieval, hybrid retrieval, and the full ranker using the same author-assigned graded relevance labels. Table 5 reports mean performance across the three main queries.

The full ranker achieves the highest Precision@5 and nDCG@5, improving mean Precision@5 to 1.000 (versus 0.733 for BM25 and 0.667 for both dense and hybrid retrieval) and improving mean nDCG@5 to 0.752 (versus 0.640 for BM25, 0.595 for dense retrieval, and 0.685 for hybrid retrieval). On the larger corpus BM25 is a stronger baseline than before, narrowing but not closing the precision gap. Because the full ranker outputs five selected papers, we

Table 5: Retriever comparison across BM25, dense retrieval, hybrid retrieval, and the full ranker. Metrics are averaged over the three main queries using author-assigned graded relevance.

System	P@5	P@10	nDCG@5	nDCG@10	R@5	R@10
BM25	0.733	0.433	0.640	0.587	0.396	0.457
Dense	0.667	0.333	0.595	0.525	0.343	0.343
Hybrid	0.667	0.500	0.685	0.658	0.343	0.520
Full ranker	1.000	–	0.752	–	0.527	–

compare it to the other retrievers primarily at @5; @10 metrics are not defined for the capped full-ranker output.

5.5 Reformulation Ablation

We tested whether the number of reformulated queries affects retrieval by re-running each main query with one, two, and four reformulations and measuring Recall@5. Mean Recall@5 was 0.493 with one reformulation, 0.527 with two, and 0.493 with four, a small benefit at two reformulations driven by the evaluation query (R2: 0.400 to 0.500), with the other two queries flat. The effect is modest: on a focused corpus the base query already retrieves much of the relevant neighborhood, so additional reformulations add little. We retain reformulation as an architectural feature whose value is more likely to appear at larger corpus scales and under the sparse-retrieval conditions where the ReAct loop activates (Section 5.7).

5.6 Schema Compliance and Failure Analysis

Two runs achieved 80% compliance (4/5) and one achieved 100% (5/5), for a mean of 86.7%. Failures occurred on papers with survey-like structure, merged contribution and novelty fields, or broader framework papers whose contribution statements did not map cleanly to the four-field schema. Structural validity of the JSON is always guaranteed by constrained decoding; the failures are field-completeness failures on atypically structured papers. The recommended fix is a fallback extraction prompt targeting such papers.

5.7 ReAct Loop Activation Under Sparse Retrieval

In the main broad-query experiments the ReAct refinement loop did not activate, because the 100-paper corpus returns sufficient candidates for any broad query on the first iteration. To test the loop directly, we constructed a sparse-retrieval stress test: five narrow queries evaluated under a strict relevance threshold that forces first-pass retrieval below the minimum-papers requirement.

Table 6 reports the result. Under the default threshold the loop remains dormant on all five queries. Under the strict threshold the loop activates on three of five queries, issuing one to two refinement iterations, and recovers to sufficient coverage (≥ 3 papers) on one of the three activations; for example, increasing retrieval from one paper to six for the paged-memory query. This demonstrates that the refinement mechanism is functional and beneficial when retrieval is sparse, while remaining correctly inactive when the corpus already supplies enough candidates.

Table 6: ReAct loop activation under a sparse-retrieval stress test. The loop fires on 3/5 queries and recovers coverage on 1/3 activations. “Iters” counts total iterations (initial pass plus refinements), so the number of refinement steps is Iters – 1.

Query (sparse)	Iter-1	Refine	Iters	Final
verbal self-reflection memory	3	no	1	3
chunked cross-attention retr.	6	no	1	6
zero-ablation attribution	0	yes	3	0
OS-style paged memory	1	yes	2	6
dialectic multi-robot col-lab.	2	yes	3	2

Table 7: Deterministic gap-matrix statistics. Empty cells are candidate structural gaps computed programmatically, not LLM-generated.

Quantity	Value
Distinct papers placed	12
Problem classes (rows)	7
Method families (columns)	10
Total cells	70
Filled cells	10
Empty cells (gaps)	60
Matrix density	0.143

5.8 Deterministic Gap Matrix Construction

The earlier LLM-inferred gap descriptions sometimes produced cross-application observations rather than clean within-corpus absence signals. We therefore construct the final gap matrix deterministically using a fixed paper-id taxonomy derived from retrieved paper metadata and contribution records. The problem $\times$ method matrix is then populated programmatically, and empty cells are reported as candidate corpus-level gaps within the retrieved paper slice. Crucially, the gap decision itself is structural: the system does not ask the LLM to directly generate the final set of gaps.

The “gaps identified” row in Table 2 refers to report-level summarized gaps, while the deterministic matrix analysis counts all 60 empty problem–method cells across the union of retrieved papers; the two are distinct metrics and are not directly comparable.

Applied to the union of papers retrieved across the three main queries, the deterministic construction yields a 7×10 matrix (7 problem classes, 10 method families) with 10 of 70 cells filled, giving a matrix density of 0.143 (Table 7). The 60 empty cells are candidate structural gaps by construction rather than generated text, making the gap set reproducible and removing the cross-application noise present in directly LLM-inferred gaps.

5.9 Gap-Matrix Validation

To assess whether deterministic empty cells correspond to plausible research gaps, we sampled 20 empty cells from the problem $\times$ method matrix and manually labeled each as *plausible*, *too*

Table 8: Validation of sampled deterministic gap-matrix empty cells.

Metric	Value
Sampled empty cells	20
Labeled cells	20
Plausible cells	12
Gap precision	0.600

broad, *indirect*, or *meaningless*. As shown in Table 8, 12 of 20 sampled empty cells were judged plausible, yielding a gap precision of 0.600.

Among the 20 sampled cells, the 8 non-plausible cases broke down as 4 indirect (only indirectly supported by the retrieved corpus), 3 too broad to be actionable, and 1 meaningless. This suggests that the deterministic matrix is useful as a corpus-level absence signal, but individual empty cells should be interpreted cautiously rather than as definitive claims about the broader literature.

5.10 Qualitative Case Study

To illustrate the type of structured output produced by the system, Table 9 shows a representative qualitative case study from the multi-agent framework query. Unlike a standard RAG summary, the system separates shared coverage, paper-specific differentiation, and corpus-level absence signals.

5.11 Baseline RAG Comparison

We implemented a baseline RAG system (retrieval followed by direct GPT-4o summarization, with no ranker, no structured extraction, and no comparison agent) and ran it on all three main queries. The baseline produces a coherent multi-paragraph summary per query in 3–4 seconds but emits no structured records, associates no overlaps with specific paper identifiers, and builds no gap matrix. The proposed system takes roughly 23 seconds (an increase of about sevenfold) but produces typed extraction records, paper-level overlaps, a structured gap matrix, and an auditable six-stage trace. Counting structured capabilities (structured contribution records, paper-level overlap IDs, per-paper differentiation, a problem $\times$ method gap matrix, and an auditable multi-stage trace), the full system supports all five while the baseline supports none.

Beyond this capability difference, we compare retrieval quality directly. Because the baseline retrieves by raw similarity order while the full system applies the ranking stage, this comparison isolates the value of ranking. Using the same author-assigned relevance labels, Table 10 reports the head-to-head: the ranker improves mean Precision@5 from 0.733 to 1.000, mean nDCG@5 from 0.548 to 0.752, and mean Recall@5 from 0.368 to 0.527. We note that the baseline’s raw-similarity retrieval is not identical to the dense retriever in Table 5: the baseline retrieves the top-15 chunks and deduplicates to papers by first (best-ranked) occurrence before truncating to five, whereas the dense retriever in Table 5 ranks at the paper level. The two therefore select different top-5 sets, which is why their @5 metrics differ. The ranking stage thus contributes measurable retrieval-quality gains on top of the shared embedding retrieval, in addition to the structured artifacts the baseline cannot produce. The

Table 9: Qualitative case study of structured comparison output for the query “Compare multi-agent LLM frameworks for collaborative reasoning.”

Query	Overlap Example	Difference Example	Gap Example
Multi-agent LLM frameworks for collaborative reasoning	AgentVerse and AutoGen both address multi-agent coordination through structured multi-agent collaboration, while AgentBench and AgentSurvey emphasize evaluation and conceptual framing.	AgentVerse focuses on emergent multi-agent collaboration; AutoGen emphasizes scalable multi-agent conversation; OpenAgents focuses on deployment infrastructure for agents in real-world settings.	Within the retrieved set, multi-agent collaboration frameworks are not paired with explicit benchmarking or deployment-oriented evaluation, suggesting an absent problem–method combination in this corpus slice.

Table 10: Retrieval-quality head-to-head: full system (with ranker) versus the baseline RAG system (raw similarity order), mean over the three main queries, author-assigned graded relevance.

System	P@5	nDCG@5	Recall@5	MRR
Baseline RAG (no ranker)	0.733	0.548	0.368	1.000
Full system (ours)	1.000	0.752	0.527	1.000

Table 11: Component ablation of structured comparison capabilities.

System Variant	Records	Overlap IDs	Diff.	Gap Matrix	Trace
Baseline RAG	No	No	No	No	No
+ Extractor	Yes	No	No	No	No
Full system	Yes	Yes	Yes	Yes	Yes

latency increase is the direct cost of structure, and is acceptable for a literature-review setting where the alternative is hours of manual reading.

5.12 Component Ablation

Isolating each stage’s contribution shows a clear progression: the baseline RAG system yields generic prose with no structure; adding the extractor yields structured four-field records but no cross-paper reasoning; and only the full system, with the comparison agent, produces paper-level overlaps, per-paper differentiation, and a gap matrix. The comparison agent is thus the component that distinguishes this system from a structured-output RAG pipeline.

5.13 External Usefulness Assessment

To obtain evaluation independent of the author, we recruited a single external rater (a graduate student familiar with LLM agents but not involved in building the system) to assess the structured output for the three main queries. The rater scored each output dimension from 1 (poor) to 5 (excellent) without access to the author’s labels, and the rater’s identity is not reported in the paper. These ratings were collected on an earlier 55-paper version of the system outputs. Because this assessment predates the final corpus expansion, we use it only as a qualitative usefulness signal, not as evidence for

Table 12: External usefulness assessment by a single independent rater (mean over the three main queries, 1–5 scale).

Dimension	Mean (1–5)
Extraction correctness	4.00
Overlap correctness	4.00
Differentiation specificity	3.00
Gap usefulness	3.00
Overall usefulness	3.33

final-system performance. Table 12 reports the mean scores across the three queries.

The rater scored extraction correctness and overlap correctness highest (4.0 each), with differentiation specificity, gap usefulness, and overall usefulness lower (3.0, 3.0, and 3.3 respectively). Two consistent critiques accompanied the scores: that per-paper differentiation describes each paper largely in isolation rather than explicitly contrasting it against the other papers in the retrieved set, and that the problem–method gaps would be more useful with deeper methodological analysis rather than high-level labels. We treat both as concrete targets for future work: the first motivating a set-aware differentiation pass, and the second a richer gap characterization. As a single-rater assessment these scores are indicative rather than conclusive, consistent with the evaluation limitation discussed in Section 7.

5.14 Cross-Model Robustness

To test whether the retrieval behavior is specific to a single LLM backbone, we re-ran the three main queries with two additional backbones, GPT-4o-mini and GPT-4.1, holding the corpus, retriever, prompts, ranking procedure, and evaluation labels fixed. Table 13 reports the mean retrieval metrics and schema compliance across the three queries.

The main structural finding is stable across backbones: all three models produce query-sensitive top-5 sets, with low mean pairwise Jaccard similarity (0.12–0.20), and all preserve the architectural capability gap over the baseline RAG system. The additional backbones also improve schema compliance from 86.7% to 100.0%. Retrieval precision and recall are lower under GPT-4o-mini and GPT-4.1. This drop partly reflects the limited coverage of the author-assigned relevance labels: the additional backbones retrieve topically related

Table 13: Cross-model robustness across three LLM backbones. Metrics are averaged over the three main queries using the same author-assigned graded relevance labels.

Backbone	P@5	nDCG@5	R@5	Schema	Jaccard
GPT-4o	1.000	0.752	0.527	86.7%	0.12
GPT-4o-mini	0.800	0.624	0.410	100.0%	0.20
GPT-4.1	0.800	0.707	0.418	100.0%	0.18

Table 14: Extended evaluation on seven additional queries (R4–R10), author-assigned graded relevance. Means for the full ten-query set (R1–R10) are shown for reference.

Query	P@5	nDCG@5	Recall@5	MRR
R4 (tool use)	1.000	0.665	0.417	1.000
R5 (memory)	1.000	0.771	0.500	1.000
R6 (planning)	0.800	0.453	0.333	1.000
R7 (RAG factuality)	1.000	0.948	0.500	1.000
R8 (single vs multi)	1.000	0.804	0.500	1.000
R9 (hallucination)	1.000	0.700	0.500	1.000
R10 (reflection)	1.000	0.791	0.556	1.000
Mean (R4–R10)	0.971	0.733	0.472	1.000
Mean (R1–R10)	0.980	0.739	0.489	1.000

papers such as AgentTuning (a corpus paper not among the cited references) and OpenAgents that were not included in the original graded-relevance set. We therefore treat the cross-model experiment as preliminary robustness evidence for query-sensitivity and schema compliance, not as a definitive model-ranking benchmark.

5.15 Extended Evaluation: Seven Additional Queries

The three main queries (R1–R3) are deliberately broad comparison queries. To probe whether the retrieval behavior holds on a wider and more varied query set, we ran seven additional queries (R4–R10) spanning tool use, agent memory, planning, RAG factuality evaluation, single- versus multi-agent architectures, hallucination reduction, and reflection. These queries use the same corpus, pipeline, and author-assigned graded-relevance protocol as the main experiments; they extend rather than replace the main results, which remain as reported above. Table 14 reports per-query retrieval metrics for R4–R10.

Two observations follow. First, query-sensitivity strengthens at the larger query set: across all ten queries no paper appears in every top-5 set, the mean pairwise Jaccard similarity is 0.115 (versus 0.12 across the three main queries), and of the 29 distinct papers selected across the 50 ranked slots, 18 are query-exclusive. The most frequently selected paper appears in only six of ten queries. This is consistent with the main finding that the ranker responds to query semantics rather than returning a fixed ordering.

Second, Precision@5 is no longer saturated on the wider set: the mean over the seven new queries is 0.971, and over all ten queries 0.980. The single sub-perfect case is R6 (planning), where an agent

evaluation benchmark was ranked into the top-5 and judged not relevant to a planning-method query under the author labels. We read this as evidence that the graded-relevance labels are discriminative rather than uniformly permissive: the same labeling protocol that yields Precision@5 of 1.000 on the main queries also penalizes a topically adjacent but off-target retrieval. Schema compliance over the ten queries averages 84.0%, close to the 86.7% reported on the three main queries. As with the main evaluation, these are single-run, author-labeled results on a 100-paper corpus and carry the same limitations discussed in Section 7.

6 Discussion

6.1 The Latency–Structure Trade-off

The defining design tension in agentic retrieval is between latency and structure. A single-call RAG baseline answers in under four seconds; our system takes roughly seven times longer. The return on that cost is a set of typed, inspectable artifacts (per-paper records, paper-level overlaps, and a gap matrix) that a downstream user or system can act on programmatically. For interactive web search this trade-off would be unacceptable; for literature review, where the user would otherwise spend hours reading and manually tracking relationships, under thirty seconds is negligible. The right operating point depends on the retrieval task, and agentic retrieval systems should make this trade-off explicit rather than defaulting to one extreme.

6.2 Query-Sensitivity as a Retrieval Property

The clearest empirical signal in this prototype evaluation is that the ranked set changes substantially with the query. Across the three main runs, no paper appears in all three top-5 sets and the mean pairwise Jaccard similarity is only 0.12. The extended seven-query evaluation strengthens this pattern: across all ten queries, no paper appears in every top-5 set, mean pairwise Jaccard similarity remains low at 0.115, and 18 of 29 distinct retrieved papers are query-exclusive. For an agentic retrieval system this is the property that justifies the architecture: if the ranked set did not vary with the query, the comparison stage would simply re-describe a fixed set of papers. Query-sensitive ranking ensures that the structured comparison is computed over a set tailored to the user’s specific question.

6.3 Structured Records as Retrieval Output

A broader implication is that retrieval output need not be passages or prose. By enforcing a schema at extraction time, the system turns each retrieved document into a typed record. This reframes retrieval as producing structured data rather than text, which in turn enables comparison operations (overlap detection, deterministic gap-matrix construction) that are awkward or impossible over free text. We see this as a useful direction for agentic IR: treat the retrieved set as a small structured database rather than a context window to be summarized.

7 Limitations

As a prototype, the system has two scope limits that frame its future work. First, the 100-paper corpus, though substantially larger

and more diverse than the initial set, remains modest relative to large-scale IR benchmarks. Second, evaluation uses author-assigned relevance labels and manually labeled gap-validation judgments, with usefulness assessed by a single external rater whose ratings were collected on an earlier 55-paper version of the outputs; additional independent raters and a broader multi-rater human study on the final outputs would further strengthen the conclusions. Scaling the corpus and expanding external evaluation are the natural next steps toward archival-grade benchmarking.

8 Conclusion

We presented the Novelty-Aware Research Agent, a prototype agentic retrieval system that layers structured multi-step reasoning on a RAG pipeline to compare research contributions across a retrieved corpus. The six-component pipeline combines query reformulation, a ReAct retrieval loop, relevance ranking, schema-guided extraction, a three-pass comparison stage with deterministic problem-method matrix construction, and answer generation. On a 100-paper corpus the system demonstrates query-sensitive retrieval: no paper appears in all three main top-5 sets, and an extended ten-query evaluation preserves this pattern with mean pairwise Jaccard similarity of 0.115 and 18 query-exclusive papers among 29 distinct retrieved papers. The system achieves 86.7% schema compliance on the three main queries and 84.0% over the ten-query set, while supporting 5/5 structured comparison capabilities versus 0/5 for a baseline RAG system. Within the three-query author-assigned evaluation, the ranker attains mean Precision@5 1.000, nDCG@5 0.752, and Recall@5 0.527, ahead of BM25, dense, and hybrid retrieval; over all ten queries, Precision@5 remains high but non-saturated at 0.980, with nDCG@5 of 0.739 and Recall@5 of 0.489. Gap-matrix validation over 20 sampled empty cells yields gap precision of 0.600. Across GPT-4o-mini and GPT-4.1, query-sensitivity remains stable and schema compliance improves, while retrieval precision is lower under the fixed labels. These results support treating retrieval output as structured data for contribution-level comparison: the payoff is a single inspectable map of what a retrieved set covers and what it leaves open, in place of a ranked list the researcher would otherwise cross-reference by hand. We leave large-scale evaluation and independent human assessment as future work.

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